



微积分第四次作业

$$\begin{aligned} 20. \lim_{x \rightarrow 3} \frac{x^2 + ax + b}{x-3} &= \lim_{x \rightarrow 3} \frac{x^2 - 6x + 9 + (a+b)x + b-9}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{(a+b)x + b-9}{x-3} = (a+b) + \lim_{x \rightarrow 3} \frac{3(a+b) + b-9}{x-3} = 2 \end{aligned}$$

$$\therefore a+b=2, 3(a+b)+b-9=0$$

$$\therefore a=-4, b=3$$

21 (5) 解: 设为 k 阶, 则 $\lim_{x \rightarrow 0} \frac{\tan^2 x - \sin^2 x}{x^k} = C$, C 为常数

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan^2 x - \sin^2 x}{x^k} &= \lim_{x \rightarrow 0} \frac{(\frac{1}{\cos^2 x} - 1) \sin^2 x}{x^k} = \lim_{x \rightarrow 0} x^2 \frac{\sin^2 x}{\cos^2 x} \\ &= \frac{x^4}{x^k} = 1 \quad (k=4 \text{ 时}) \end{aligned}$$

$\therefore$ 关于 x 的阶为 4。

(6) 解: 设为 k 阶, 则 $\lim_{x \rightarrow 0} \frac{\cos mx - \cos nx}{x^k} = C$, C 为常数

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos mx - \cos nx}{x^k} &= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{m+n}{2} x \cdot \sin \frac{m-n}{2} x}{x^k} \\ &= \frac{n^2 - m^2}{2} \frac{x^2}{x^k} = \frac{n^2 - m^2}{2} \quad (k=2 \text{ 时}) \end{aligned}$$

$\therefore$ 关于 x 的阶为 2。

22. (3) 解: 设 $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^k} = C$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^k} &= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x + \cos^2 x)}{x^k} = \lim_{x \rightarrow 0} \frac{3(1 - \cos x)}{x^k} \\ &= \lim_{x \rightarrow 0} \frac{3 \cdot \frac{x^2}{2}}{x^k} = \frac{3}{2} \quad (k=2 \text{ 时}) \end{aligned}$$

$\therefore$ 主部是 $\frac{3}{2}x^2$

(4) 解: 设 $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x^5} - \sqrt{1-2x^5}}{x^k} = C$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{1+2x^5} - \sqrt{1-2x^5}}{x^k} &= \lim_{x \rightarrow 0} \frac{1}{x^k} \frac{4x^5}{\sqrt{1+2x^5} + \sqrt{1-2x^5}} = \lim_{x \rightarrow 0} \frac{2x^5}{x^k} \\ &= 2 \quad (k=5 \text{ 时}) \end{aligned}$$

$\therefore$ 主部是 $2x^5$

1.3.1 解: 已知 $\cos x + a$, 0 , $bx^2 + c$ 各三段本身连续

$\therefore$ 只需要考虑断点处的连续性

$$\therefore \text{需让 } \lim_{x \rightarrow 0} f(x) = 0$$

$$\lim_{x \rightarrow 0} (\cos x + a) = 1 + a = 0 \Rightarrow a = -1,$$

$$\lim_{x \rightarrow 0} (bx^2 + c) = c = 0 \Rightarrow c = 0$$

$\therefore a = -1, c = 0, b$ 任意实数

2(3) 解: $f(x) = (1+ax)^{\frac{1}{x}}$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} [(1+ax)^{\frac{1}{ax}}]^a = e^a$$

$\therefore x=0$ 时, 令 $f(x) = e^a$

14) 解: $f(x) = e^{-\frac{1}{x^2}}$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} e^{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} e^{-\frac{1}{x^2}} = 0$$

$\therefore x=0$ 时, 令 $f(x) = 0$